Lab 3 – Microstrip Stub Tuning

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Pre-Lab

Microstrip is a PCB transmission line that supports quasi-TEM waves and offers small size, high bandwidth, and low dispersion, making it widely used in microwave circuits. This lab follows the design and simulation of shunt open-circuited single and double-stub tuning networks on a 50-Ω microstrip to achieve impedance matching.

Obj 1

With a substrate thickness of 1.6 mm made of FR4 Epoxy ($\varepsilon_r = 4.4$) and a target characteristic impedance of 50 Ω, the microstrip trace width was designed to be 3.1 mm using MATLAB. The guided wavelength was calculated to be 0.0548 m at the design frequency of 3 GHz. The return loss with a 200 Ω termination was calculated to be 4.437 dB.

(Obj 1) $Z_0 = 50 \quad \epsilon_r = 4.4 \quad d = 0.0016\,m \quad f = 3e9$

$$A = \frac{Z_0}{60}\sqrt{\frac{\epsilon_r+1}{2} + \frac{\epsilon_r-1}{\epsilon_r+1}\left(0.231 - \frac{0.11}{\epsilon_r}\right)} = 1.5299 \qquad \frac{W}{d} = \frac{8e^{A}}{e^{2A}-2} = 1.9119 < 2 \checkmark$$

$$W = \frac{W}{d}\cdot d = \boxed{0.31\,cm} \qquad \epsilon_e = \frac{\epsilon_r+1}{2} + \frac{\epsilon_r-1}{2}\cdot\frac{1}{\sqrt{1+12d/W}} = 3.33$$

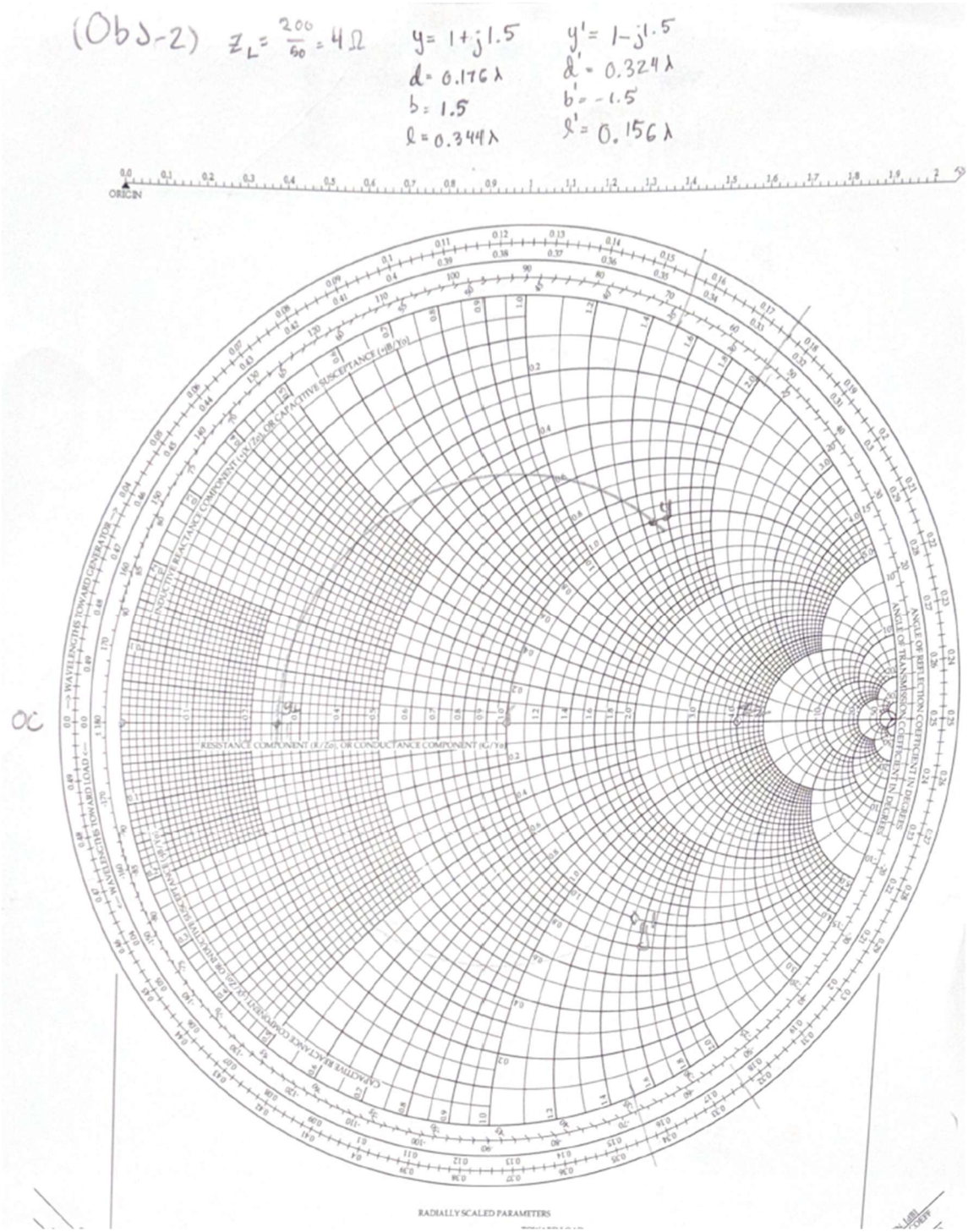
$$\beta = \sqrt{\epsilon_e}\cdot\frac{\omega}{c_0} = 114.66$$

$$RL = -20\log|\Gamma| = \boxed{4.437\,dB} \qquad \lambda = \frac{2\pi}{\beta} = \boxed{0.0548\,m}$$

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{150}{250} = 0.6$$

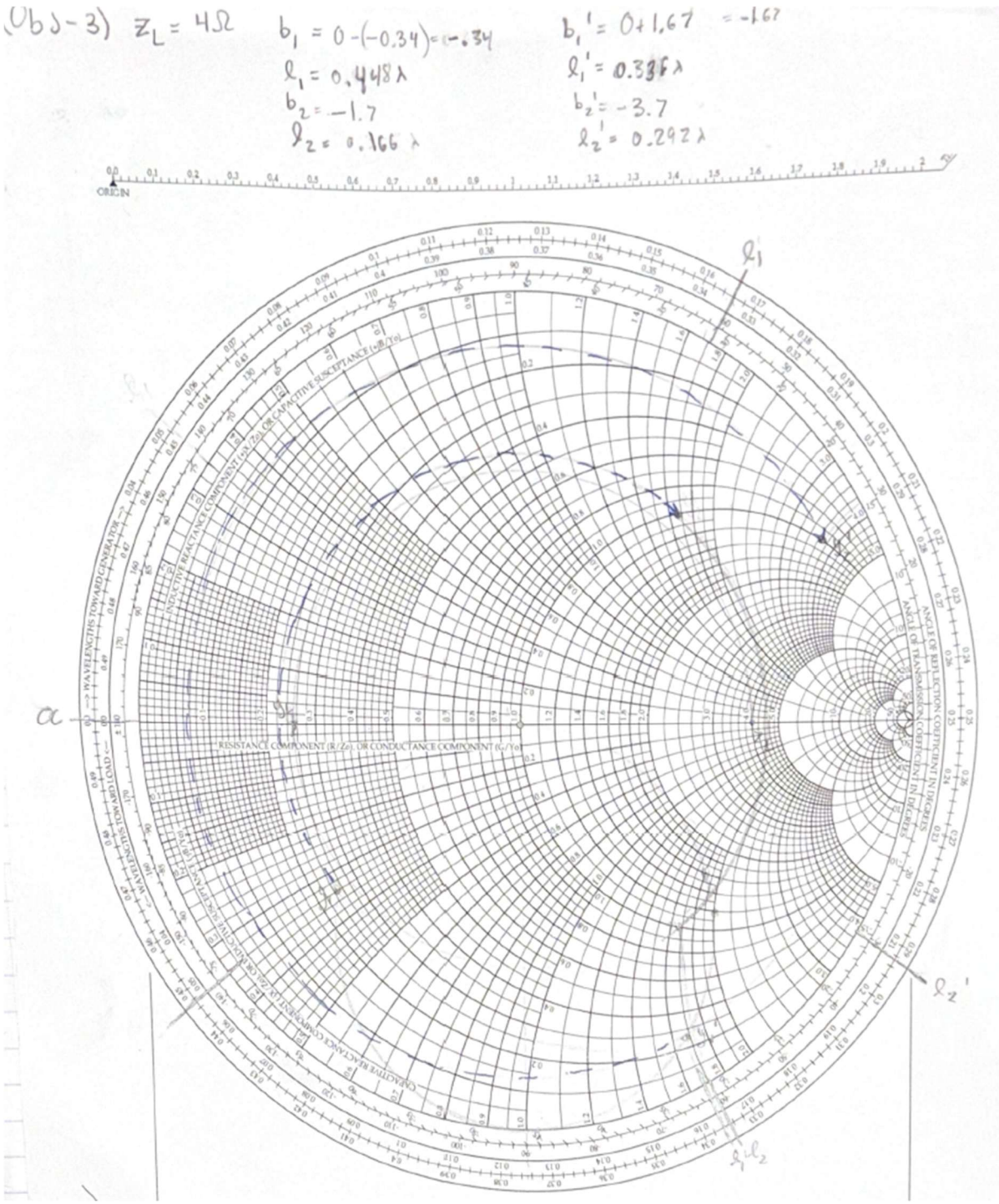
Obj 2

The shunt, open-circuit, single-stub tuning was designed using a Smith chart. The first solution was implemented in part 2, setting $d = .176\lambda$ and $l = .344\lambda$, where λ is the guided wavelength at 3 GHz, 0.0548m.



Obj 3

The shunt, open-circuit, double-stub tuning was also designed using Smith chart. Both solutions were implemented and tested in part 3.



Lab

Part 1 — 50Ω Microstrip Line (Driven Terminal + Lumped Ports)

Model Setup

Figure 1 shows the model setup; a simple PCB composed of a FR4 Epoxy substrate of thickness 1.6 mm, and a single microstrip PEC trace of width 3.1 mm and 0 thickness, as designed in Obj-1. The board dimensions are 1.5 λ_0 by 2 λ_0 and the model is surrounded by an airbox with radiation boundary condition. The trace sits in the center of the board, spanning the length of the longer side, with a PEC ground plane defined on the opposite face of the substrate.

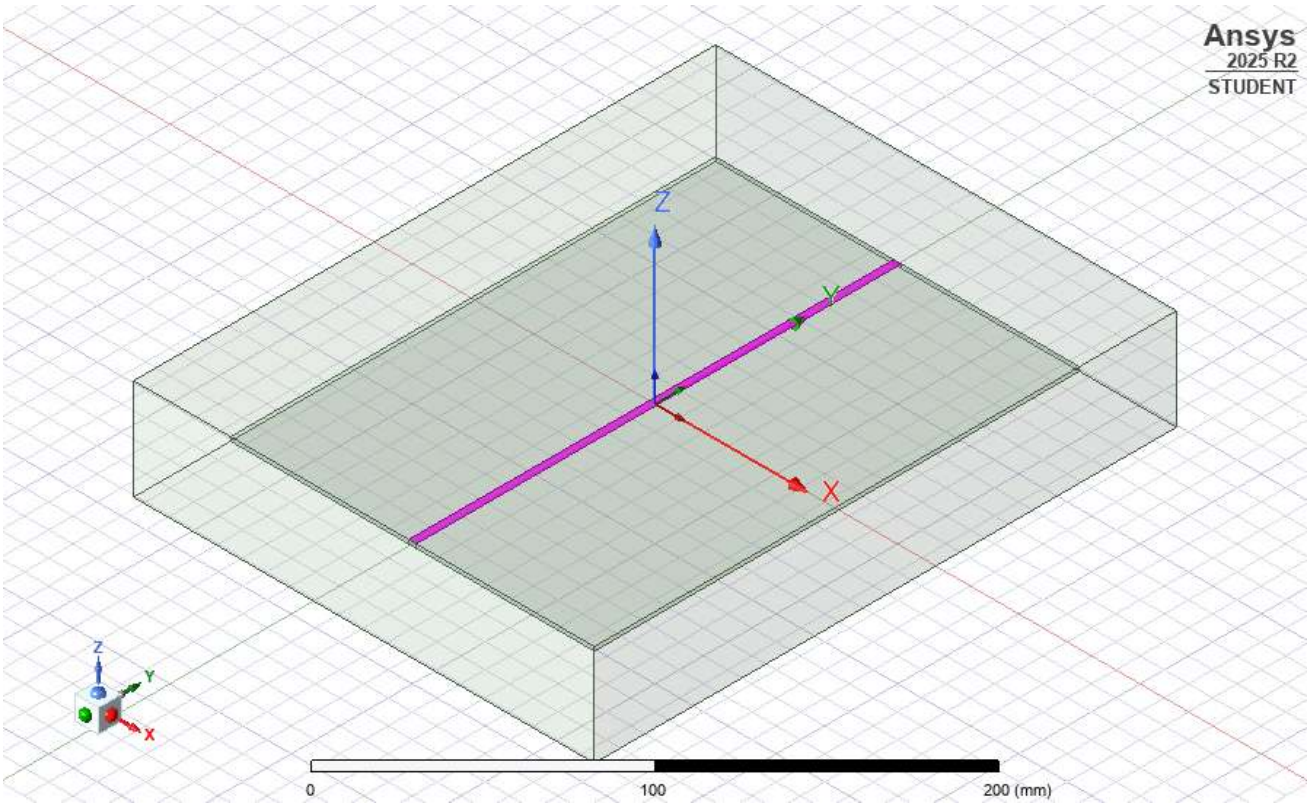


Figure 1: Device model for microstrip line

Terminal Lumped ports were set at each end as shown in Figures 2 and 3. Both ports were renormalized to 50Ω. Only the dominant mode was simulated.

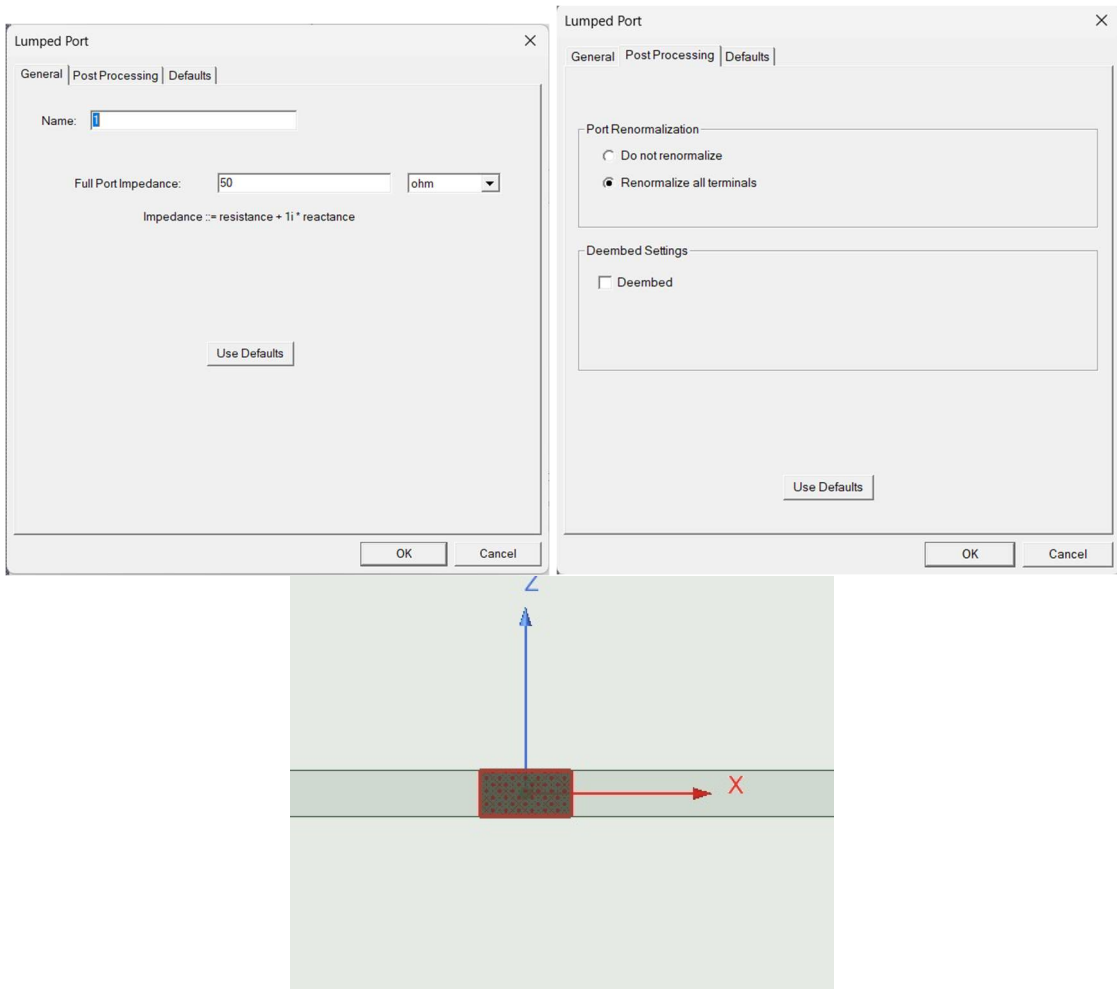


Figure 2: Port 1 setup for basic microstrip line

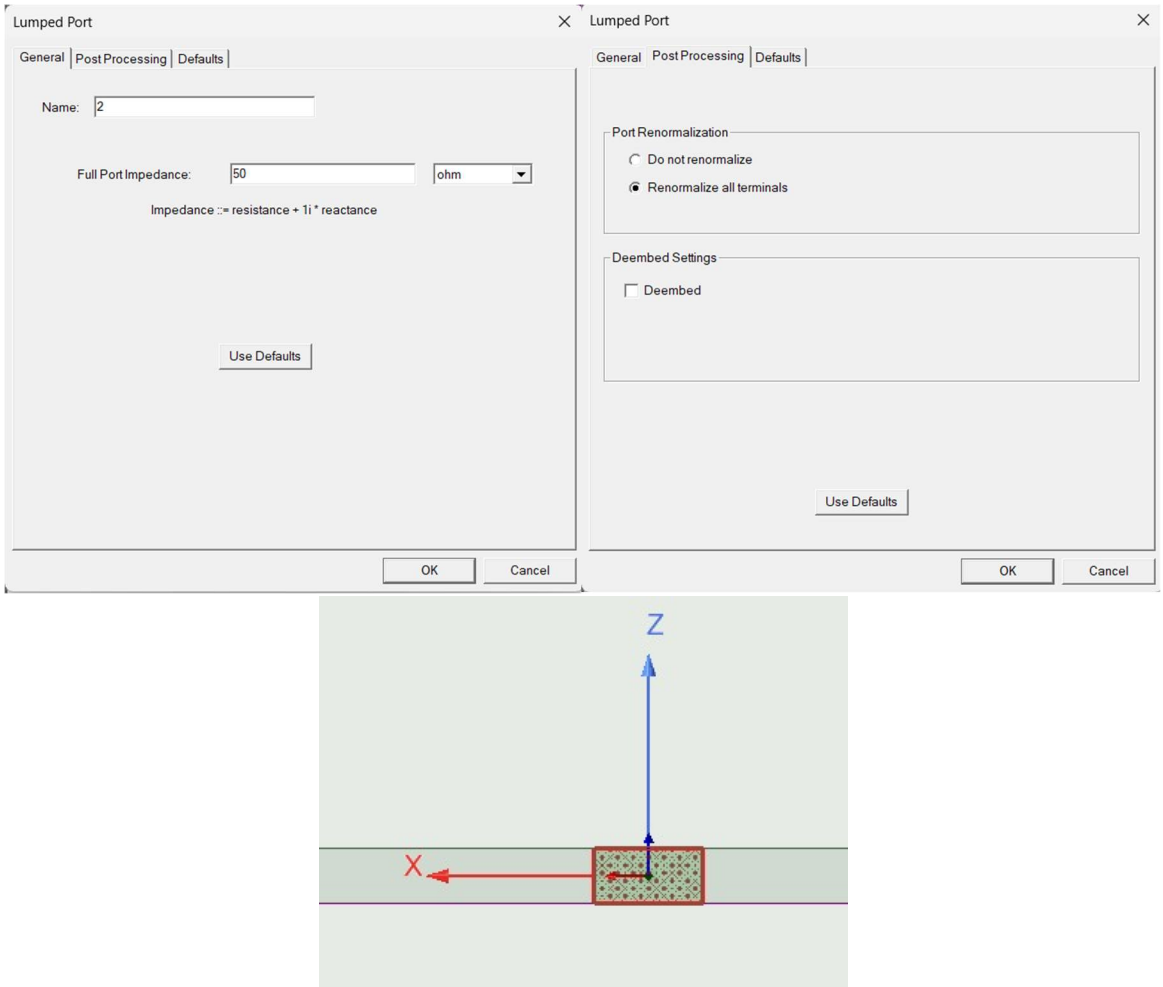
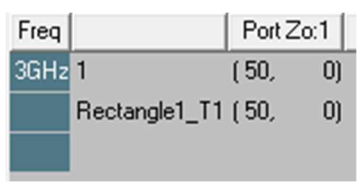


Figure 3: Port 2 setup for basic microstrip line

(Res-1) Shown in Figure 4, the simulated characteristic impedance perfectly matches the designed value.



Freq		Port Zo:1
3GHz	1	(50, 0)
	Rectangle1_T1	(50, 0)

Figure 4: Characteristic impedance for basic microstrip line

(Res-2) Figure 5 shows the plot of S(1,1). While the minimum reflection occurs slightly higher than the design frequency, reflection remains low across the operating range, as expected with matched terminations.

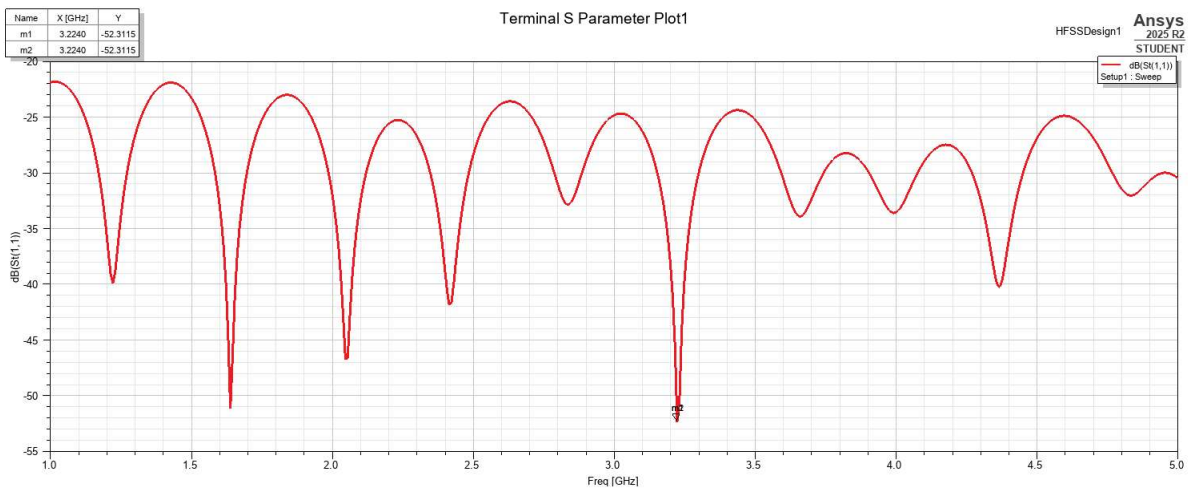


Figure 5: S(1,1) operating band sweep for basic microstrip line

Renormalized Port-1

Using the same model and port setup, port 1 is renormalized to 200Ω to match the design as shown in Figure 6

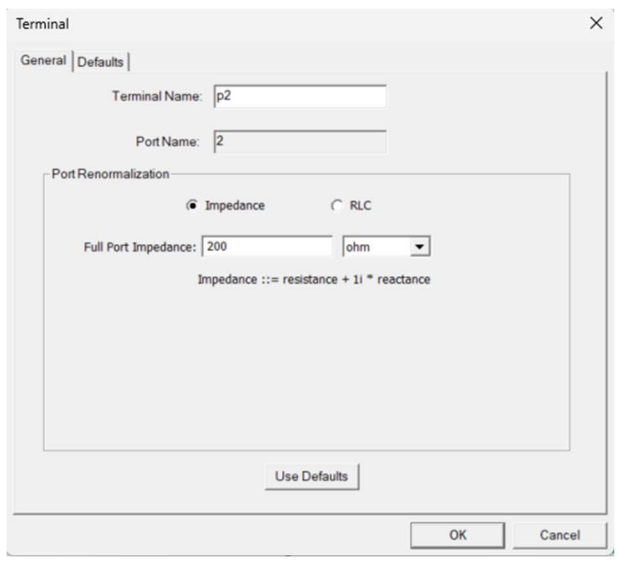


Figure 6: Port-1 renormalization to 200Ω

(Res-3) Figure 7 shows the plot of S(1,1) after renormalizing to 200Ω. The return loss is slightly higher than the expected value of 4.4 dB. This discrepancy can be attributed to approximation and simulation errors. However, the significantly higher reflection compared to the previous result with matched ports aligns with expected behavior.

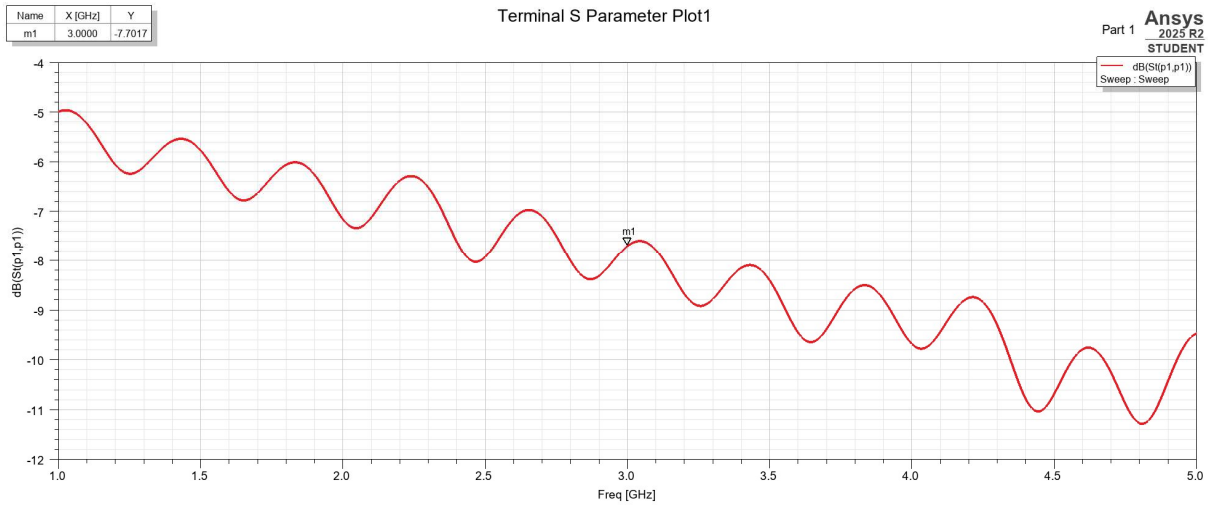


Figure 7: S(1,1) Operating band sweep for microstrip line with 200Ω Termination

Part 2 Shunt, Open-Circuit, Single-Stub Tuning Network

The model shown in Figure 8 was used for all part 2 simulations. The setup is the same as part 1, with port 2 replaced by a Lumped-RLC boundary condition with impedance 200Ω, and a shunt, open-circuit, single stub added as designed in Obj-2, with $d = .176\lambda$ and $l = .344\lambda$, where λ is the guided wavelength at 3 GHz, 0.0548m.

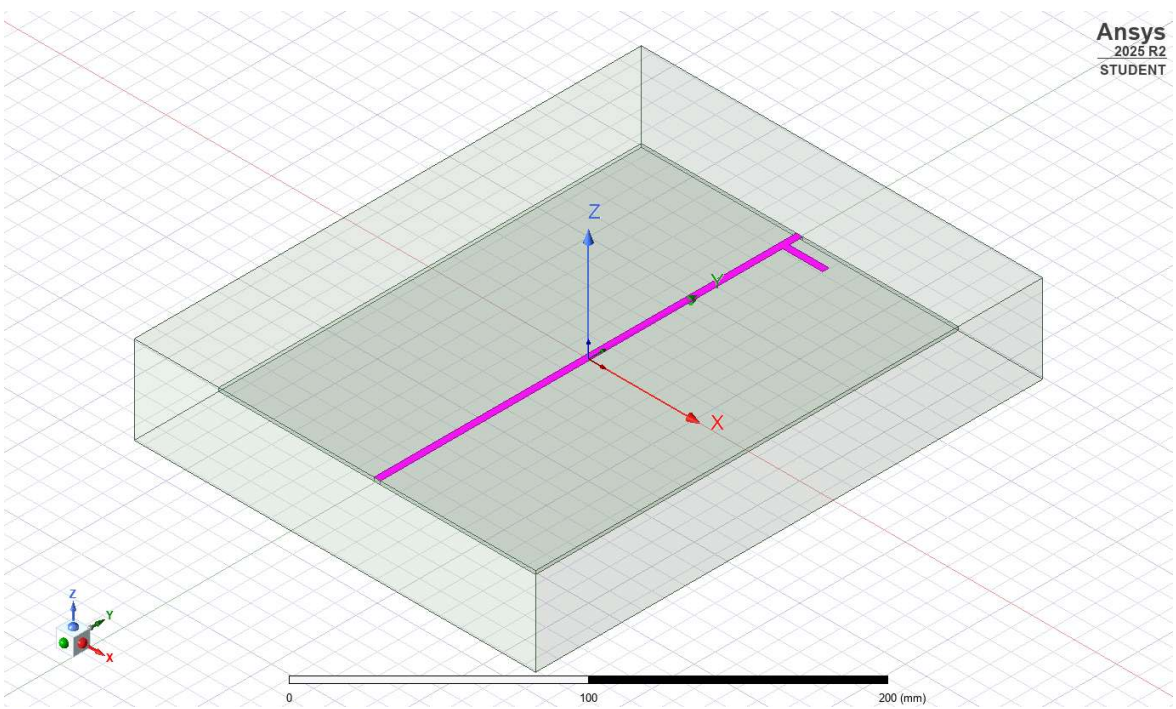


Figure 8: Device model for shunt, OC, single-stub tuning network

The current flow line and setup for the Lumped-RLC boundary condition is shown in Figure 9.

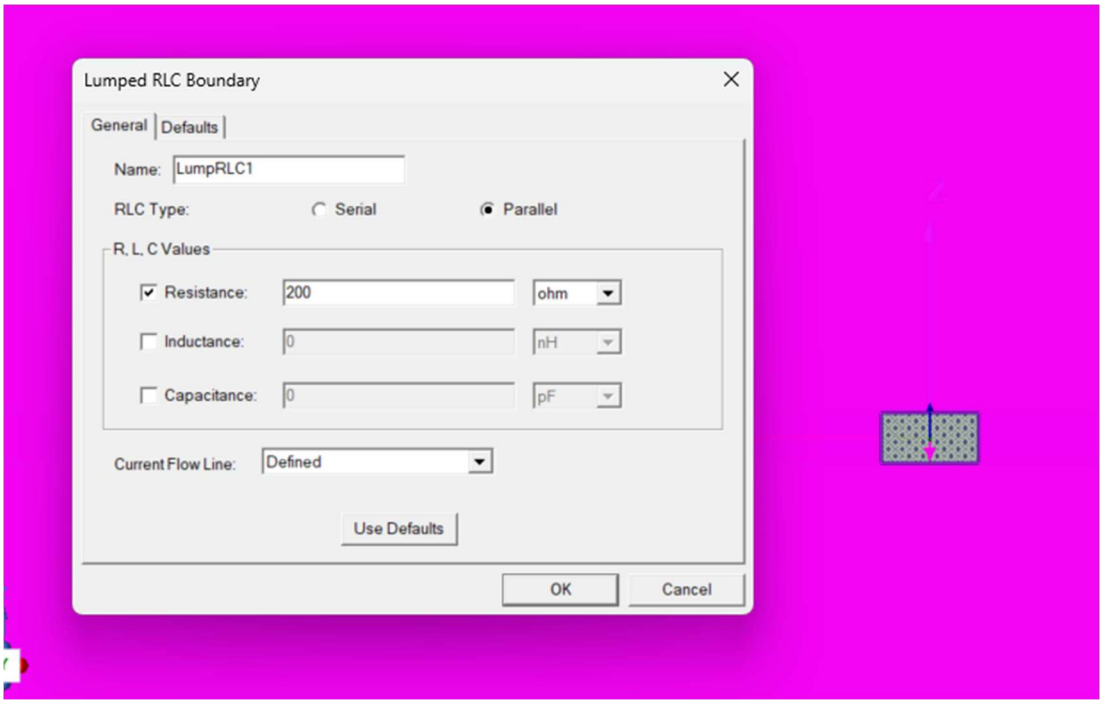


Figure 9: Lumped-RLC boundary condition for port

(Res-4) Figure 10 shows the S(1,1) plot for the single-stub tuning network. The minimum reflection occurs very close to the design frequency, and the -10 dB matching bandwidth is ~ 1 GHz. When compared to Res-3, it is evident that the single-stub design achieved successful impedance matching at and around the design frequency, since the reflection is significantly reduced.

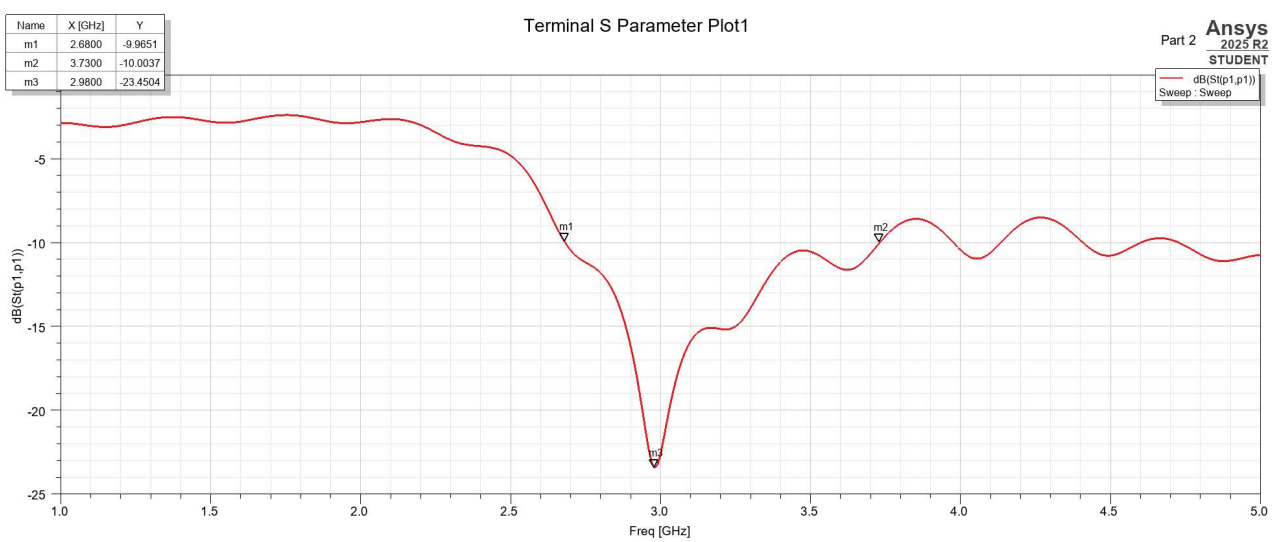


Figure 10: S(1,1) Operating band sweep for shunt, OC, single-stub tuning network

Part 3 Shunt, Open-Circuit, Double-Stub Tuning Network

The model shown in Figure 11 was used for all part 3 simulations. The setup is the same as part 2, with a second stub added. Using the same λ, the guided wavelength at design frequency, the first stub is placed a

half-wavelength away from port 2, and the second stub is placed another 3/8 wavelength down the line. The lengths of each stub are varied to test the two designs from Obj-3.

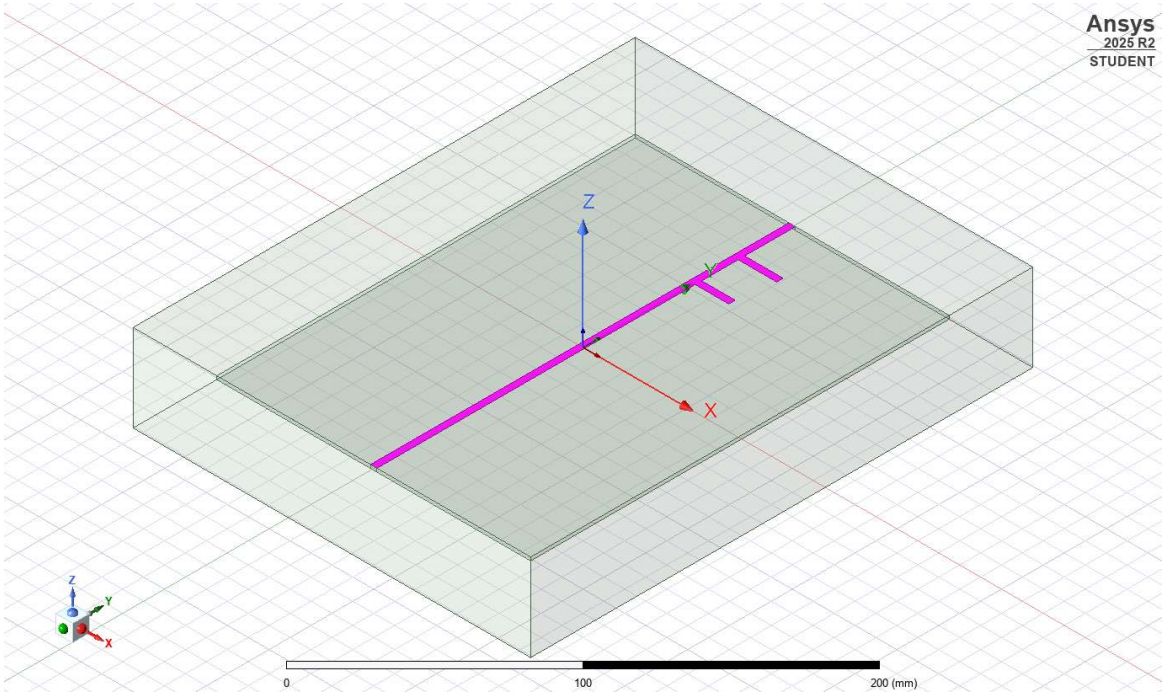


Figure 11: Device model for shunt, OC, double-stub tuning network

The first solution set uses $l_1 = .336\lambda$ and $l_2 = .293\lambda$.

(Res-5) Figure 12 shows the S(1,1) plot for the double-stub tuning network. The minimum reflection occurs close to the design frequency, but the -10 dB matching bandwidth is only ~ 60 MHz, much narrower than the single-stub design. As a result, the reflection at the design frequency is actually worse than the basic strip with no matching. This may be due to smith chart errors or simulation innaccuracies since it does not align with theoretical expectations.

Figure 12: S(1,1) Operating band sweep for shunt, OC, double-stub tuning network (soln. 1)

(Res-6) Figure 13 shows the corresponding Smith chart for the S(1,1) operating band sweep of the tuning network. The markers show the frequencies of the closest gamma magnitudes corresponding to -10 dB ($|\Gamma| \approx 0.316$). These frequencies closely match the -10dB frequencies in the rectangular plot from Figure 12.

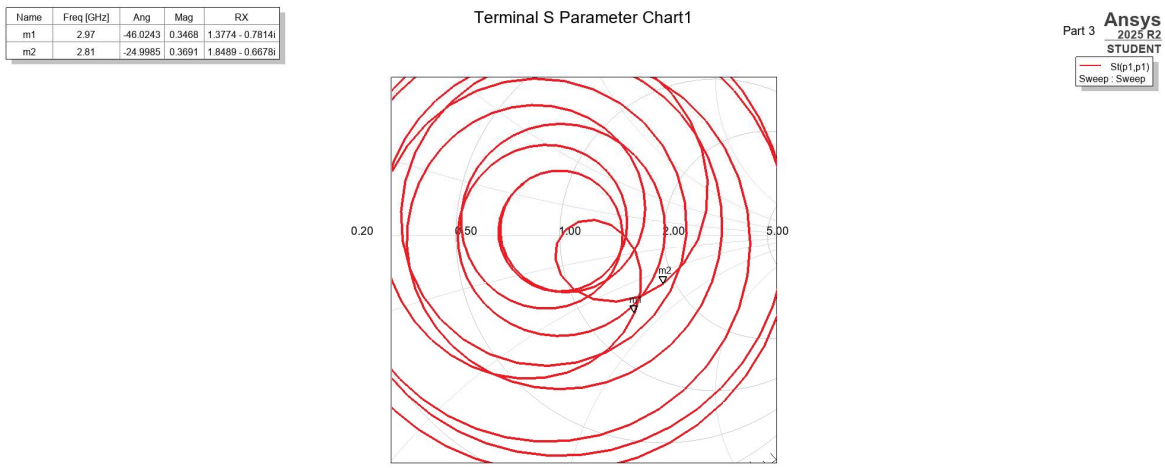


Figure 13: S(1,1) Smith chart for shunt, OC, double-stub tuning network (soln. 1)

The second solution set uses $l_1 = .448\lambda$ and $l_2 = .166\lambda$.

(Res-7) Figure 14 shows the S(1,1) plot for the double-stub tuning network, second solution. The minimum reflection again occurs close to the design frequency. For this solution, the matching bandwidth -10 dB matching bandwidth is larger than the previous one, ~ 190 MHz, but still narrower than the single-stub design. However, this matching band does include the design frequency, with a RL of 16.8 dB at 3 GHz. Therefore this solution achieves impedance matching at and around the design frequency.

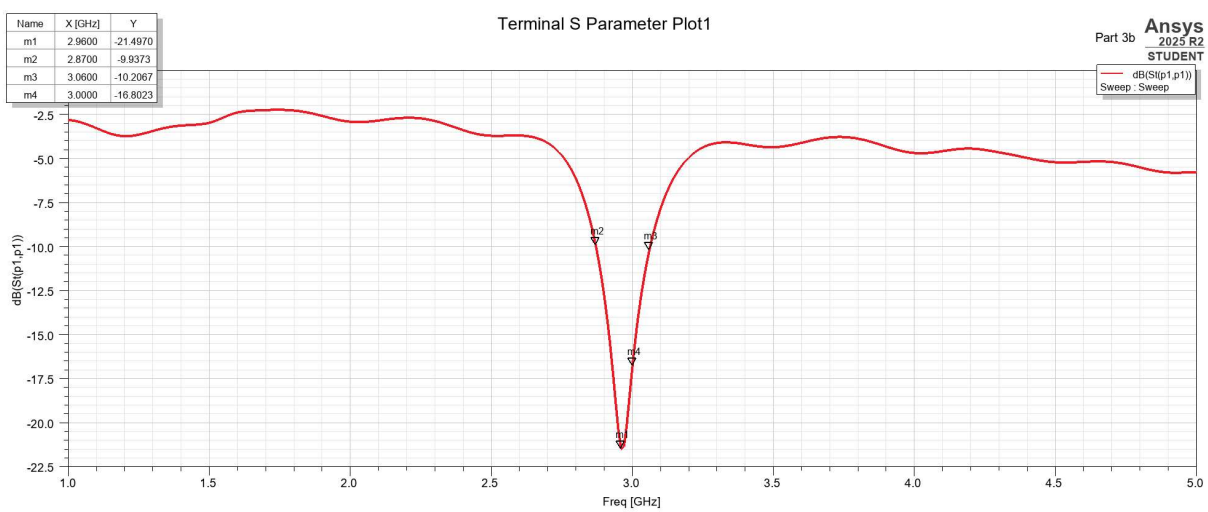


Figure 14: S(1,1) Operating band sweep for shunt, OC, double-stub tuning network (soln. 2)

(Res-8) Figure 15 shows the corresponding Smith chart for the S(1,1) operating band sweep of the tuning network. Again, with markers at ($|\Gamma| \approx 0.316$), these frequencies closely match the -10dB frequencies in the rectangular plot from Figure 14.

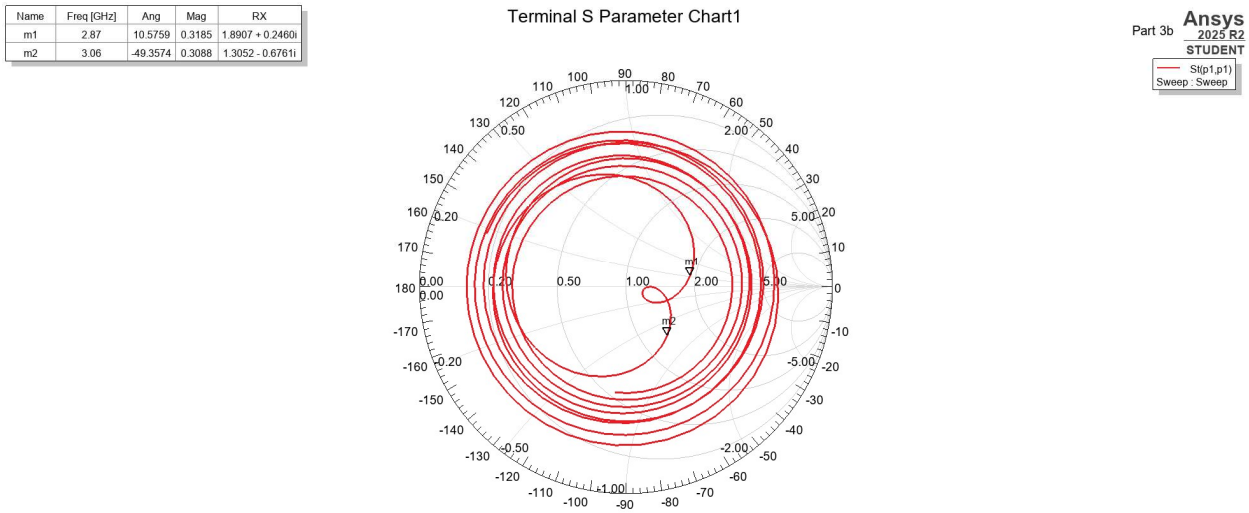


Figure 15: S(1,1) Smith chart for shunt, OC, double-stub tuning network (soln. 2)

Report Checklist

1. I have included screenshots of all the HFSS models. **AM**
2. I have included screenshots of all port setups, including integration lines where appropriate. **AM**
3. I have included all necessary results in graphs, tables, or statements. **AM**
4. I have interpreted and commented on all my results. **AM**